consideration to be a phantom problem. The matter is simplest when an error is lurking in the assumptions. In this case, of course, it is immaterial whether the erroneous assumption was introduced deliberately or has just escaped detection. A lucid example is the famous problem of perpetual motion, i.e. the problem of devising a periodically functioning apparatus which will perform mechanical work perpetually without any other change in nature. Since the existence of such an apparatus would contradict the principle of the conservation of energy, such an apparatus cannot possibly occur in nature, so that this problem is a phantom problem. Of course, one may raise the following argument: “The principle of the conservation of energy, after all, is an experimental law. Accordingly, although today it is considered to be universal and all-embracing, its validity may one day have to be restricted—and in fact, such a curtailment of its universal applicability has been sometimes suspected in nuclear physics—and the problem of perpetual motion would then suddenly become genuine. Its meaninglessness is, therefore, by no means absolute.”

This counter-argument may actually acquire practical significance, as is demonstrated especially clearly by the example of a no less well-known problem in chemistry: The ancient problem of changing base metal, for instance, mercury, into gold. Originally, prior to the birth of a scientific chemistry, this problem was considered to be pregnant with portentious meaning, and many a learned—and unlearned—mind was zealously occupied with it. But later, as the theory of chemical elements was developed and became universally accepted, the transmutation of metals turned into a phantom problem. In recent times, since the discovery of artificial radioactivity, the situation has again been reversed. The fact is that today it no longer seems to be fundamentally impossible to discover a process for removing a proton from the nucleus of the mercury atom and an electron from its shell. This operation would change the mercury atom into a gold atom. Therefore, at the present stage of science, the ancient quest of the alchemists no longer belongs to the class of phantom problems.

However, these examples must by no means be construed as indicating that the meaninglessness of a phantom problem is never absolute, but simply dependent on whether or not a certain theory is accepted as valid. There are also many phantom problems which are indubitably doomed to remain such forever. One of these, for instance, is the problem which used to keep many a great physicist busy for many years: The study of the mechanical properties of the luminiferous ether. The meaninglessness of this problem follows from its basic premise, which postulates that light vibrations are of a mechanical nature. This premise is erroneous, and must so remain forever.

Here is another example, taken from the field of physiology: It is a well known fact that the convex lens of the human eye projects an inverted image